# Prapti Trivedi

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## WORK EXPERIENCE

**Amazon (Amazon Music)** *Sunnyvale, USA*
*Applied Scientist* *May 2025 – Present*

- Developed a **generative music retrieval** method pairing hierarchical RQ-VAE multimodal tokens with an instruction-tuned LLM that adapts modality weights to query intent, improving Recall@100 by 7% over baselines.
- Developed a multi-agent catalog framework for **200M+ tracks**, integrating agent consensus and metadata validation for defect detection, self-correction, and active learning, with 95% precision in genre, era, mood, and activity classification.
- Built a **podcast intelligence pipeline** combining transcript entity extraction, entity linking, knowledge graphs, and graph-enhanced RAG driving a 2% lift in listening time in online A/B tests.
- Developed a multimodal, multi-tower classification model to identify **non-music content** under a business-specific definition across **15M tracks**, with 98% precision at 80% recall enabling automated $10M royalty savings.

**Collinear AI** *Mountain View, USA*
*Machine Learning Engineer* (*Founding Team*) *July 2024 – April 2025*

- Developed state-of-the-art LLM-as-a-judge model for **Safety** with **Self-rationalization** and **Iterative DPO** for fine-grained customizable evaluation criteria, outperforming other post-training methods and baseline models by 6%.
- Developed a unified platform for **synthetic data curation** with **low-latency Reward Models** and LLM-as-Judges for **Instruction Following, Coherence, Reliability, and Complexity**, increasing downstream model accuracy by 9%.
- Created character AI personas by **Knowledge Infusion** and improved **RAG**, followed by custom auto-alignment with high-quality synthetic data curation, resulting in a 34% improvement in custom evaluation scores.

**Seagate** *Longmont, USA*
*Data Science Intern* *May 2023 - August 2023*

- Finetuned various base LLMs using **LoRA and QLoRA** with **recommendation-driven and engagement-guided prompting** to give personalised recommendations for hard-drives and SSD; leading to a 15% increase in **Conversion Rate** in purchases.
- Spearheaded the **Text-to-SQL** initiative for conversational database interactions via LLMs; optimized SQL generation by fine-tuning, few-shot learning, and prompt engineering, resulting in a 33% reduction in engineering hours.

**Guise AI Inc.** *Missouri, USA*
*Machine Learning Engineer* *April 2021 - July 2022*

- Developed a people-count and dwell-time analytics application with enhanced object detection and tracking algorithm, for analyzing real-time customer engagement attributes, boosting customer retention by 28%.
- Designed an end-to-end model optimization pipeline integrating low bit quantization, Knowledge Distillation, optimizing hardware operations and **increased FPS** by 20% in all use cases, on low compute **edge devices**.
- **Incorporated** continual learning(AutoML) for all use cases, making models robust to Out Of Distribution data **on the fly**; increased the accuracy on average by 32.3% and reducing development-to-production time by 46%.

**Sprinklr** *Gurugram, India*
*Data Scientist* *July 2020 - March 2021*

- Developed an end-to-end ASR model for downstream CSM, implementing Word Pair Encoding multilingual transliteration schemes and self-supervised learning decreasing WER by 27% and **reducing the dependency** on large-scale labelled data.
- Decreased the **Agent Handoff rate** by 33% for a Voice Chat-bot(a virtual agent for banking) serving **low-resource** Indic languages, by designing a custom code-switching(auto language detection) **conversation flow architecture**.

## EDUCATION

**New York University – Courant Institute of Mathematical Sciences**
*M.S. in Computer Science* *GPA: 3.84/4.00*

**Dhirubhai Ambani Institute of Information and Communication Technology**
*B.Tech. in Information and Communication Technology* *GPA: 8.54/10*

## TECHNICAL SKILLS

**Languages**: Python, C, C++, SQL, Scala, Java
**Tools & Technologies**: PyTorch, TensorFlow, Hugging Face TRL, Transformers, vLLM, TensorRT, Metaflow, Spark, Kubernetes, Flask, MongoDB, Docker, AWS, GCP, Git, Linux, Kafka, Tableau, YARN, Hive, TVM, CUDA, Databricks, MLflow

## PUBLICATIONS

**Cats Confuse Reasoning LLM: Query Agnostic Adversarial Triggers for Reasoning Models**
*Meghana Rajeev, Rajkumar Ramamurthy, Prapti Trivedi, Vikas Yadav, Oluwanifemi Bamgbose, Sathwik Tejaswi* ***COLM 25***

**Self-Rationalization improves LLM as a finegrained judge**
***Prapti Trivedi, Aditya Gulati, Oliver Molenschot****, Meghana Rajeev, James Zou* ***ACL ARR(In Review)***

**Bayesian ResNet for Indic handwritten digit recognition**
*Purva Mhasakar,* ***Prapti Trivedi****, Srimanta Mandal, Suman K. Mitra* ***SN Computer Science 2021***

**Multichannel CNN for Facial Expression Recognition**
***Prapti Trivedi****, Purva Mhasakar, Sujata, Suman K. Mitra* ***PReMI 2019***